A direct Witt-vector proof for $G = \mathbb{Z}/p^r$

the symmetric power kills every ramification break below d, carries and all

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Claim. **??** follows directly, for every $G = \mathbb{Z}/p^r$, by repeating the proven $\mathbb{Z}/p$ computation (`RamificationAfterBlowupIsTame.tex`, lines 508–575) with *Witt vectors* in place of a single Artin–Schreier equation. The inductive route of `a_wrong_path.tex` is bypassed entirely: there is no base change along q, hence no absorption to rule out. The mechanism that defeated the boundary $d = r$ (the Witt carry) reappears here as an explicit term whose pole is removed by the symmetric power precisely when the ramification bound holds.

1 The datum of the symmetric power is a Witt sum

Reduce as in the thesis to $k = \bar{k}$ and $\mathcal{P} \to U$ totally ramified at p_∞ (`RamificationAfterBlowupIsTame.tex`, line 646). Locally $\widehat{\mathcal{O}}_{C,p_\infty} = k[[x]]$, and $\mathcal{P}$ is the Artin–Schreier–Witt W_r-torsor

$$\wp(\vec{Y}) = \vec{a}, \qquad \vec{a} = (a_0, \ldots, a_{r-1}) \in W_r\big(k((x))\big), \qquad \wp := F - \mathrm{id},$$

where F is the Witt Frobenius. Normalise $\vec{a}$ to its reduced form, so a_l has pole order m_l at $x = 0$ (prime to p, or a_l regular). At the exceptional-divisor point V_C of the blow-up of $C^{(d)}$ at $d \cdot p_\infty$, the completed field is $\widehat{K_C} = \kappa_C((e_d))$ with $e_d = \prod_i x_i$ a uniformiser, $\kappa_C = k(e_1/e_d, \ldots, e_{d-1}/e_d)$, and (the thesis's blow-up model)

$$v_C(e_i) = 1 \ \ (1 \le i \le d), \qquad v_C(e_i/e_j) = 0. \tag{1}$$

Lemma 1 (the symmetric-power datum). *Restricted to $\widehat{K_C}$, the G-cover $\mathcal{P}^{[d]_G}$ is the W_r-Artin–Schreier–Witt extension*

$$\wp(\vec{Y}) = \vec{\alpha}, \qquad \vec{\alpha} = \sum_{i=1}^{d}{}^{W} \ \vec{a}(x_i) \ \in W_r\big(\widehat{K_C}\big),$$

the d-fold Witt sum *of the local data at the d colliding points.*

Proof. This is the W_r-analogue of the thesis computation (`RamificationAfterBlowupIsTame.tex`, lines 521–547). The cover $\mathcal{P}^{[d]_G}$ is $\mathcal{P}^{\times d}/(K_G \rtimes S_d)$ with $K_G = \ker(\Sigma : G^d \to G)$. On generators, G^d

acts by Witt translation $\vec{X}_i \mapsto \vec{X}_i + \vec{g}_i$, and the Witt sum $\vec{Y} := \sum_i^W \vec{X}_i$ satisfies, for $(\vec{g}_i) \in K_G$ (i.e. $\sum_i^W \vec{g}_i = 0$),

$$\vec{Y} \;\longmapsto\; \sum_i^W (\vec{X}_i + \vec{g}_i) = \vec{Y} + \sum_i^W \vec{g}_i = \vec{Y},$$

so $\vec{Y}$ is K_G-invariant; and since $\wp$ is a homomorphism of Witt vectors,

$$\wp(\vec{Y}) = \sum_i^W \wp(\vec{X}_i) = \sum_i^W \vec{a}(x_i) = \vec{\alpha}.$$

The extension $k((x_\bullet))(\vec{Y})/k((x_\bullet))$ has degree $p^r = |G|$, which equals $[\,(\,\cdot\,)^{K_G} : R^{\otimes d}] = |G|^d/|K_G|$; hence $R^{\otimes d}[\vec{Y}] = (S^{\otimes d})^{K_G}$. Taking S_d-invariants (the Witt sum $\vec{\alpha}$ is symmetric in the x_i) gives $\mathcal{P}^{[d]_G}$ over $K(C^{(d)})$, with datum $\vec{\alpha}$. Completing at V_C gives the statement. □

2 Isobaric weight bounds the pole by the break

Write $\alpha_j = S_j\big(\vec{a}(x_1), \ldots, \vec{a}(x_d)\big)$, where S_j is the j-th universal Witt-sum polynomial. Set $y_i := 1/x_i$ and call the largest pole order of a symmetric Laurent expression F at the collision $x_\bullet = 0$ its *y-degree* $\deg_y F$ (the top total degree in the y_i).

Lemma 2 (isobaric pole bound). *S_j is isobaric of weight p^j, where the level-l variable carries weight p^l. Consequently*

$$\deg_y \alpha_j \;\le\; \max_{0\le l\le j} p^{j-l}\, m_l \;=:\; u_{j+1}, \tag{2}$$

the $(j{+}1)$-th upper ramification break of $\mathcal{P}$ at p_∞.

Proof. Isobaricity is the defining property of the Witt-sum polynomials (from $w_n = \sum_{l\le n} p^l X_l^{p^{n-l}}$, each w_n is isobaric of weight p^n, and S_j is built from the w_n, $n \le j$). A monomial of S_j is a product $\prod_t X_{l_t}^{(i_t)}$ with $\sum_t p^{l_t} = p^j$. Substituting $X_l^{(i)} \mapsto a_l(x_i)$, of pole order m_l, the monomial has y-degree $\sum_t m_{l_t}$. Maximising $\sum_t m_{l_t}$ subject to $\sum_t p^{l_t} = p^j$ is a linear program whose optimum is at a pure vertex (p^{j-l} factors all of one level l), giving $\max_l p^{j-l} m_l$. The last equality is the Schmid–Witt/Brylinski break formula for Artin–Schreier–Witt extensions ([Tho05] for the $\mathbb{Z}/p$ layer; the general statement is classical). □

Remark. Inequality (2) is in fact an *equality* (the optimal pure term survives mod p); see Section 6. Only "$\le$" is needed below.

3 Low-degree symmetric functions are integral at V_C

The following generalises the thesis lemma (`RamificationAfterBlowupIsTame.tex`, lines 549–572) from power sums $\sum_i f(x_i)$ to *all* symmetric functions — which is what Lemma 1 produces.

Lemma 3 (integrality). *Let $F \in k[x_1^{\pm 1}, \ldots, x_d^{\pm 1}]^{S_d}$ be symmetric with $\deg_y F < d$. Then $v_C(F) \ge 0$; in fact $F \in \kappa_C$.*

Proof. Decompose $F = \sum_D F_D$ into parts F_D homogeneous of y-degree $D < d$ (parts of y-degree ≤ 0 are polynomials in the x_i, hence in $k[e_1, \ldots, e_d]$, of $v_C \ge 0$). A homogeneous symmetric polynomial of degree D in $y_1, \dots, y_d$ is a polynomial in the elementary symmetric functions $e_1(y), \dots, e_D(y)$ — only these, since $e_k(y)$ has degree k and $D < d$ forbids $e_d(y)$. Using $e_k(y) = e_{d-k}(x)/e_d(x)$ with $1 \le d-k \le d-1$, every monomial $\prod_k e_k(y)^{c_k}$ ($\sum_k kc_k = D$) becomes

$$\frac{\prod_k e_{d-k}(x)^{c_k}}{e_d(x)^{\sum_k c_k}}, \qquad v_C = \left(\sum_k c_k\right) \cdot 1 - \left(\sum_k c_k\right) \cdot 1 = 0$$

by (1). Hence $v_C(F_D) \ge 0$, and $F \in \kappa_C$. □

4 The theorem

Theorem 4. *Let $\mathcal{P} \to U$ be a totally ramified $\mathbb{Z}/p^r$-torsor at p_∞ with ramification bounded by d (i.e. the top upper break $u_r < d$, equivalently $\max_l p^{r-1-l} m_l < d$). Then $\mathcal{P}^{[d]_G}$ is unramified at η_C.*

Proof. By Lemma 1, $\mathcal{P}^{[d]_G}|_{\widehat{K_C}}$ is $\wp(\vec{Y}) = \vec{\alpha}$ with $\vec{\alpha} = \sum_i^W \vec{a}(x_i)$. By Lemma 2, for each $0 \le j \le r-1$,

$$\deg_y \alpha_j \;\le\; u_{j+1} \;\le\; u_r \;<\; d.$$

By Lemma 3, $\alpha_j \in \kappa_C \subseteq \mathcal{O}_{V_C}$ for all j, so $\vec{\alpha} \in W_r(\mathcal{O}_{V_C})$. As $\wp : \mathbb{W}_r \to \mathbb{W}_r$ is étale (it is $F - \mathrm{id}$, with $d\wp = -\mathrm{id}$ invertible), an Artin–Schreier–Witt extension with *integral* datum is unramified. Hence E/K_C — i.e. $\mathcal{P}^{[d]_G} \to U^{(d)}$ — is unramified at $V_C = \eta_C$. □

After the (already established) reductions to $G = \mathbb{Z}/p^r$ totally ramified, this is exactly **??**; the general finite abelian case follows by the structure theorem and **??** as in the thesis.

5 The sharp form: pole degree = break, and the boundary

Equality in (2) gives the precise picture: the j-th Witt component of the symmetric-power datum has pole order *exactly* the break u_{j+1}, and

$$\mathcal{P}^{[d]_G} \text{ unramified at } \eta_C \iff u_r < d \iff \text{the bound.}$$

This explains, uniformly, the boundary failures recorded elsewhere:

- $p = 2$, $r = 2$, $m_0 = 1$: $u_2 = 2$. The carry $\alpha_1 \supseteq e_2(x^{-1})$ is $1/e_2$ (pole) at $d = 2 = u_2$ but e_1/e_3 (unit) at $d = 3 > u_2$ — matching `first_case_r2_d3.tex` and the counterexample of **??**.
- $p = 3$, $r = 2$, $m_0 = 1$: $u_2 = 3$. Ramified at $d = 3$ ($u_2 = d$, bound fails), unramified at $d = 4$ (Section 6). Note $r < d$ *holds* at $d = 3$ yet the conclusion fails: the correct hypothesis is the bound $u_r < d$, of which $r < d$ is only the weaker shadow (Lemma 2 forces $u_r \ge r$ in the honest case).

6 Computational verification and open points

The companion scripts implement the Witt sum (via ghost components), compute α_j explicitly, and read off $\deg_y$ and v_C.

- `witt_general.py`: $r = 2$. Confirms $\deg_y \alpha_1 = \max(p\, m_0, m_1) = u_2$ *exactly* for $p \in \{2, 3, 5\}$ and several (m_0, m_1); the test case $p = 3, r = 2, d = 4$ is unramified ($v_C(\alpha_0) = v_C(\alpha_1) = 0$); the boundary $p = 3, d = 3$ is ramified ($v_C(\alpha_1) = -1$).

- `witt_r3.py`: $r = 3$. Confirms $\deg_y \alpha_j = u_{j+1} = \max_l p^{j-l} m_l$ (the nested carries), hence unramifiedness for $d > u_3$.

What still needs the author's scrutiny.

1. **Lemma 1** (the structural heart): that $\mathcal{P}^{[d]_G}|_{\widehat{K_C}}$ is given by the Witt sum $\sum_i^W \vec{a}(x_i)$. The argument mirrors the thesis's $\mathbb{Z}/p$ case line-for-line, but the K_G-invariant / degree count and the passage to the completion should be checked with the same care the thesis spends on **??**.

2. **Reduced representative.** Lemma 2 uses the reduced Witt datum (pole orders m_l realising the breaks). One should confirm $\mathcal{P}$ may be so normalised over $k((x))$ before forming $\vec{\alpha}$ (standard, and done for $\mathbb{Z}/p$ in [Tho05]).

3. **Break formula.** The identification $u_{j+1} = \max_l p^{j-l} m_l$ is classical for Artin–Schreier–Witt; cite it precisely (e.g. Garuti/Brylinski/Kato) where the thesis cites [Tho05] for $r = 1$. Only the inequality $\leq$ enters Theorem 4, so even a one-sided bound suffices.

Bottom line. Once the bound $u_r < d$ is fed in, the symmetric power makes the entire Artin–Schreier–Witt datum integral at η_C, carries included — a direct proof, free of the absorption obstruction that the inductive route could not clear. The wrong path was wrong only because it tried to reconstruct the top break across the base change q; computed in one shot inside the symmetric power, every break below d simply disappears.

References

[Tho05] Lara Thomas. "Ramification groups in Artin-Schreier-Witt extensions". In: *Journal de théorie des nombres de Bordeaux* 17.2 (2005), pp. 689–720. DOI: `10.5802/jtnb.514`. URL: `https://jtnb.centre-mersenne.org/articles/10.5802/jtnb.514/`.